

## Introduction

QueryBot is an AI-powered web application that enables users to interact with databases using plain English, eliminating the need for SQL knowledge. It integrates Gemini AI with a live PostgreSQL backend to deliver instant, accurate query results and exportable reports.

## Methodology

The system fetches the live database schema at runtime and injects it into a structured Gemini AI prompt, ensuring all generated SQL uses exact table and column names. Each query undergoes safety validation before execution, and results are processed into structured query and PDF reports automatically.

## System Architecture

The user sends a natural language request through the Presentation Layer to the backend, where the Text Processing Module validates it before passing it to the Gemini AI engine. Gemini generates SQL using the live database schema fetched from PostgreSQL; the query is validated and executed, and the results are returned as a structured Query and PDF report through the API to the user.

## Experiments

QueryBot was tested on a PostgreSQL law firm database across tables using 12 natural language queries, ranging from single-table selections to multi-table joins, aggregations, and edge cases. Schema-aware prompts achieved accurate execution consistently.

## Conclusion

QueryBot demonstrates that natural language database interaction is achievable without sacrificing accuracy or security. By combining dynamic schema awareness, AI-driven query generation, and automated reporting, the system successfully bridges the gap between non-technical users and structured data.

QueryBot

image-logo

Fill the following to connect to the database:

Server: e.g. localhost or 192.168.1.45

Database: e.g. hr\_database

Username: e.g. Ahmed

Port: default port number: 5432

Password:

Connect

Any sensitive data won't be stored and only used during active sessions.

QueryBot

Database Connected 15:05

How can I help you?

User

List all lawyers and their hourly rates, sorted from most to least expensive

QueryBot

Query returned 4 record(s).

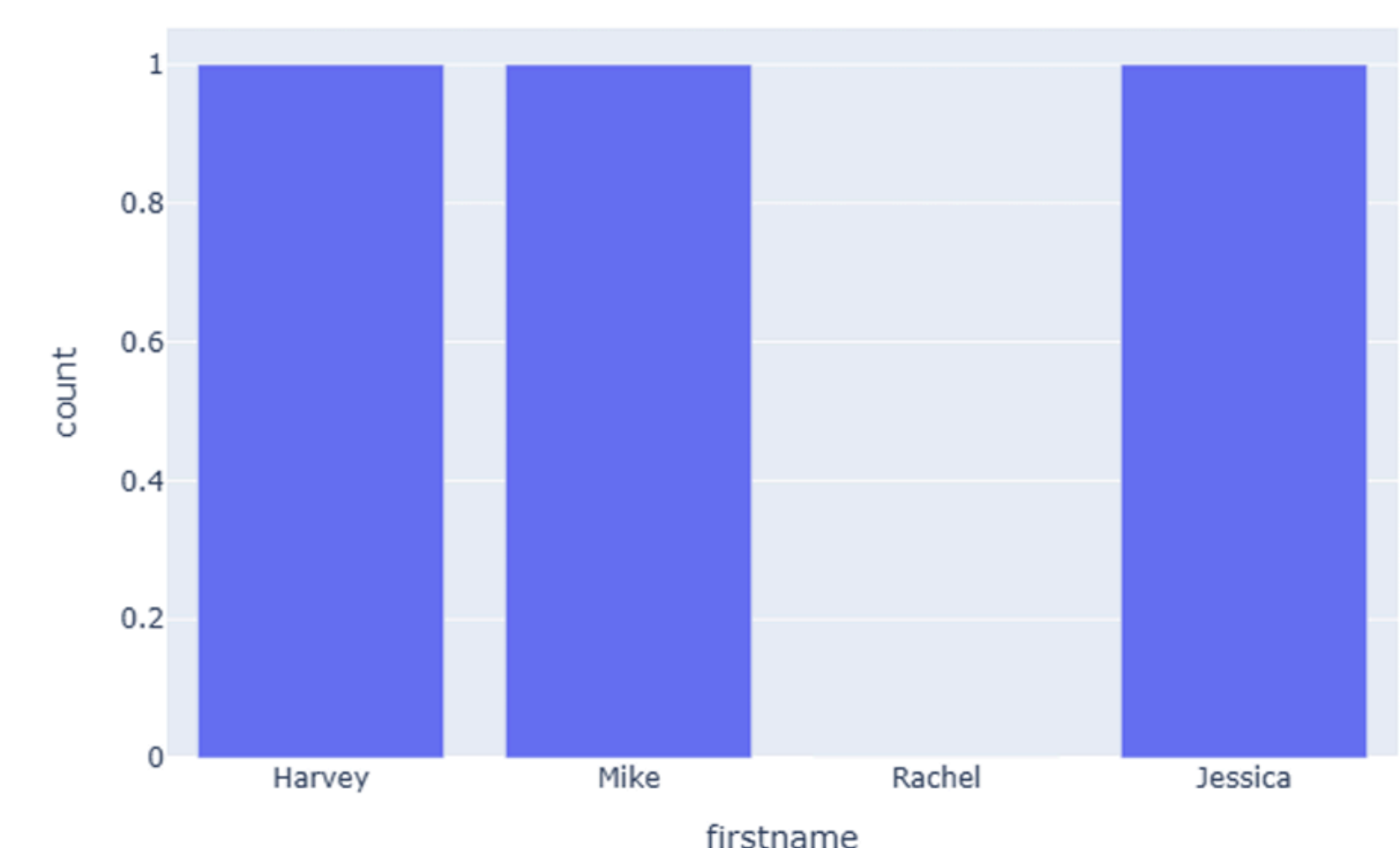
| firstname | lastname | hourlyrate |
|-----------|----------|------------|
| Jessica   | Pearson  | 900        |
| Harvey    | Specter  | 850        |
| Mike      | Ross     | 450        |
| Rachel    | Zane     | 400        |

### QueryBot Report

Summary: Query returned 4 record(s).

Total Rows: 4

Query returned 4 record(s).



### Data Preview:

| firstname | lastname | count |
|-----------|----------|-------|
| Harvey    | Specter  | 1     |
| Mike      | Ross     | 1     |
| Rachel    | Zane     | 0     |
| Jessica   | Pearson  | 1     |

### Students:

|                              |           |
|------------------------------|-----------|
| Amirah Khaled Alharbi        | 421202265 |
| Raghad Hussain Alharbi       | 422205803 |
| Ahood Awaad Aljabri          | 412205506 |
| Nouf Ayesb Aljabri           | 431215251 |
| Fatima Abdulrahman Almutairi | 391202849 |
| Khuzama Fahhad Alharbi       | 431201846 |